RANGE

Range and codomain, side by side

Where outputs are allowed to land, and where they actually land.

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COMPARED

	CODOMAIN	RANGE
WHAT IT IS	the set where outputs are declared to live	the set of outputs actually produced
FIXED BY	the definition $f : A \to B$	the rule of the function
SQUARING ON THE REALS	$\mathbb{R}$	$[0, \infty)$
	Negative numbers are in the codomain but never appear as outputs.	
CUBING ON THE REALS	$\mathbb{R}$	$\mathbb{R}$
	Every real number is the cube of some real number, so the range fills the codomain: the function is onto.	
IN ELEMENTARY ALGEBRA	usually all real numbers, by default	the set that is actually asked for
	The range is always a subset of the codomain; the two are equal exactly when the function is onto.	